



Project Phase 1

Requirements Elicitation and Management

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Introduction

In the rapidly evolving landscape of healthcare, integrating technology is crucial to meet the increasing demand for accessible, efficient, and personalized patient care. This project emerged from the need to adapt to changes brought about by the COVID-19 pandemic, heightened demand for healthcare services, and the swift advancement of technology. These factors highlighted the necessity of enhancing management systems, particularly customer relationship management (CRM) systems, to be more flexible, modern, efficient, and responsive.

This project aims to develop a comprehensive Hospital Customer Relationship Management (CRM) system tailored to the specific needs of a healthcare institution [1]. At this stage, our primary goal is to gather and understand the detailed requirements of the system's users, including patients, medical professionals, and hospital staff, and to seek additional information as needed to complete these requirements and identify key actors.

From customer insights and feedback, it is clear that there is a strong desire for an application that makes accessing healthcare services faster and more convenient, while also improving safety and the relationship between patients and the healthcare organization. To address these needs, we are developing a new, comprehensive application designed to enhance the efficiency of healthcare access and foster better patient engagement with the organization. Our approach begins with gathering and defining essential functional and non-functional requirements.

Functional requirements for the system include capabilities for remote consultations, automated reminders, secure telemedicine through video consultations, and medication schedule management. Additionally, the system will support secure payment processing for medical services, provide communication channels directly with healthcare providers, and facilitate medication delivery. The system will also include secure data storage, support multiple languages, and offer detailed reporting on user activity.

Non-functional requirements are also essential to ensure the system's overall effectiveness and user satisfaction. The CRM system must have a user-friendly interface to minimize learning curves and accommodate all users, including elderly or disabled patients. Accessibility features will include support for screen readers, adjustable text sizes, and voice

control. The system should be highly reliable, with all-day availability and minimal downtime, supported by a scalable cloud-based infrastructure. Cross-platform compatibility for both iOS and Android devices is necessary to provide a seamless user experience. Moreover, robust security measures are crucial for protecting sensitive patient data. The system must include multi-factor authentication, advanced encryption with AES-256, and regular security audits. To manage threats effectively, advanced security tools such as Endpoint Detection and Response (EDR) [2], Security Information and Event Management (SIEM) [3], and Security Orchestration, Automation, and Response (SOAR) will be implemented [4]. These tools will aid in detecting, analyzing, and responding to security incidents promptly. Additionally, the system must comply with key healthcare regulations, including the PDPA [5], Thai Medical Act [6], and HIPAA, to ensure legal and ethical handling of patient information.

In summary, the development of this Hospital CRM system is a strategic response to the evolving needs of the healthcare industry. By focusing on user-friendly design, robust security, and comprehensive functionality, this project aims to deliver a solution that enhances patient satisfaction, improves operational efficiency, and supports the digital transformation of healthcare services. All in all, these planning and defining the objectives and direction for this phase of the project will serve as the critical framework and structure for smooth transitioning into the next stages of the project.

Functional & Non-functional Requirements

1. Appointment Management

- **Functional requirement:**

FR1: The system shall allow patients to make remote appointments.

FR2: The system shall send reminders to patients about their appointments.

FR3: The system shall allow patients to manage their appointment schedules.

FR4: The system shall provide a search function to help patients find doctors, appointments, and medical records.

- **Non-functional requirement:**

NFR1: The system must be available 24/7, utilizing a cloud-based architecture to ensure high availability and minimal downtime.

NFR2: The system should support up to 10,000 concurrent transactions. Implement load balancing techniques and closed-based architecture to manage the high user load. Consider using microservices for scalability.

NFR3: The system should be easy to use and not require extensive training. The user interface (UI) design must consider the needs of elderly patients, ensuring a seamless and intuitive user experience.

NFR4: The system should meet ISO 27001 standards for information security management.

2. Telemedicine

- **Functional requirement:**

FR5: The system shall support telemedicine, including video consultations, dosage reminders and symptom checks.

FR6: The system shall allow doctors to issue new prescriptions via the app.

FR7: The system shall allow patients to consult with doctors via video calls, chat, or in-person appointments.

FR8: The system shall facilitate the delivery of prescribed medications to patients' homes

- **Non-functional requirement:** -

3. Notifications and Communication

- **Functional requirement:**

FR9:

The system shall enable patients to access health tips,

The system shall enable patients to view hospital operating hours

The system shall enable patients to stay updated with hospital news.

The system shall enable patients to receive important notifications.

FR10: The system shall notify patients when their medication dosage is complete.

FR11: The system shall enable direct communication between patients and hospital staff for support and inquiries.

FR12: The system shall support the distribution of promotional content to patients.

- **Non-functional requirement:**

NFR5: The system must protect patient data through multi-factor authentication (MFA) and encryption standards such as AES-256 for data at rest and in transit.

4. Security and Compliance

- **Functional requirement:**

FR13: The system shall securely store patient personal information, including medical records and appointment history.

FR14: The system shall allow doctors to update patient records post-consultation.

FR15: The system shall allow patients to view and manage their medical history.

FR16: The system shall allow patients to access services in multiple languages.

FR17: The system shall process payments for medical services and medications securely.

FR18: The system shall provide secure user authentication and role-based access control.

- **Non-functional requirement:**

NFR6: The system must comply with relevant regulations, including the Personal Data Protection Act (PDPA) [5], the Thai Medical Act [6], and, if applicable, the Health Insurance Portability and Accountability Act (HIPAA) for international patients.

NFR7: Regular security audits and penetration testing should be conducted at least quarterly to identify and address vulnerabilities.

NFR8: The system should implement advanced security tools like Endpoint Detection and Response (EDR) [2], Security Information and Event Management (SIEM) [3], and Security Orchestration, Automation, and Response (SOAR) to manage threats effectively [4].

NFR9: The system should be designed with accessibility features, including support for color blindness.

NFR10: The system should include an ethical framework to ensure patient data is used responsibly and transparently.

5. Payment and Billing

- **Functional requirement:**

FR19: The system must support credit card payment systems and mobile banking.

FR20: The system should be able to create detailed invoices or bills for services rendered and medications dispensed.

- **Non-functional requirement: -**

6. User Experience

- **Functional requirement: -**

- **Non-functional requirement:**

NFR11: The system should be compatible with both iOS and Android operating systems, ensuring a consistent user experience across different devices.

7. System Maintenance and Integration

- **Functional requirement:**

FR21: The system shall generate monthly and annual reports on user activity.

FR22: The system shall connect to the database of the hospital.

FR23: The system shall allow for the integration of third-party health applications and devices.

- **Non-functional requirement: -**

Total:

Functional requirement: 23 requirements

Non-functional requirement: 11 requirements

Actors

Primary Actors

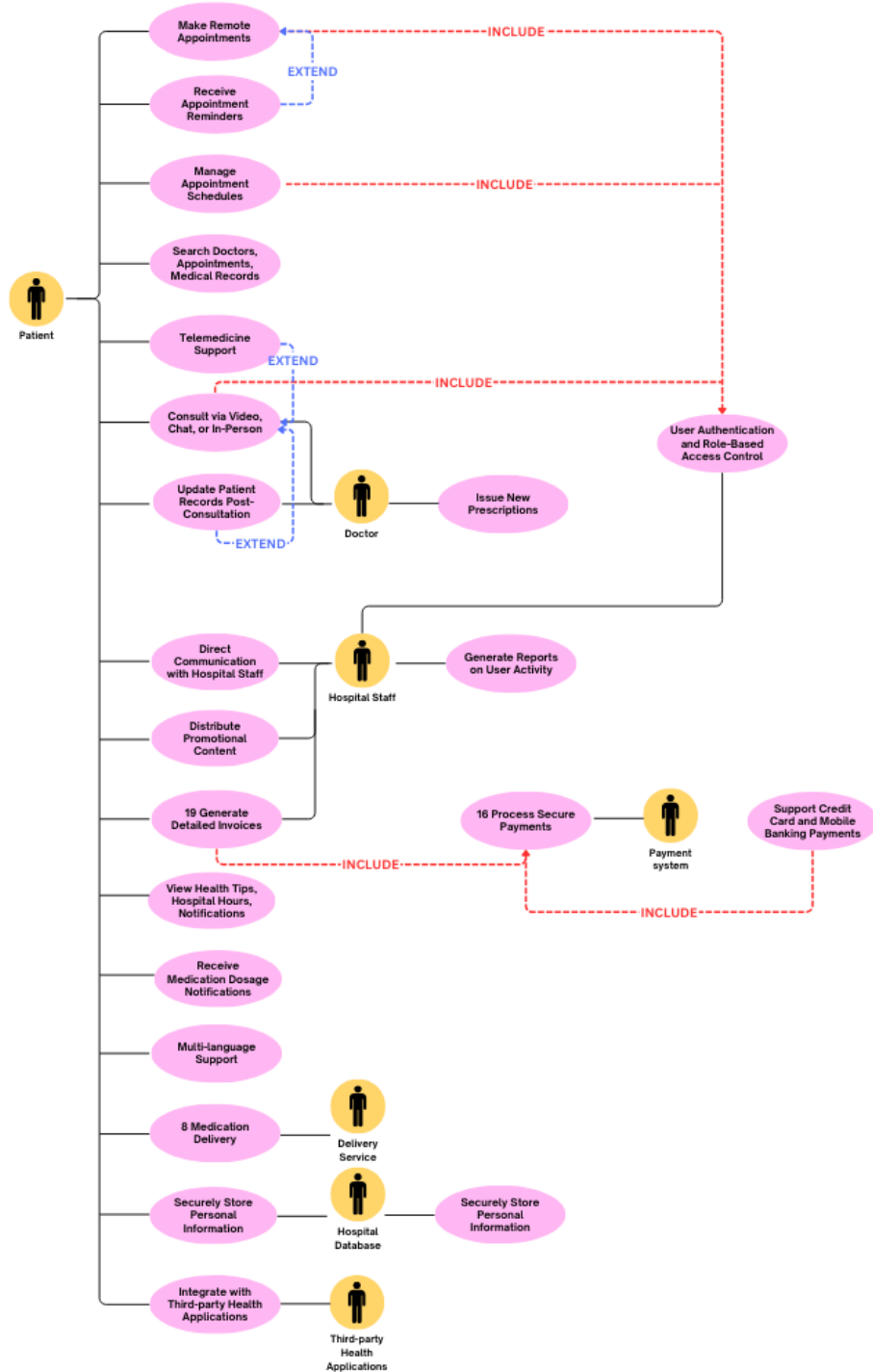
- **Patient/User:** The primary users of the system. They can create accounts, book appointments, manage their schedules, participate in telemedicine consultations, track medication schedules, and provide feedback on their experiences. This group includes patients with diverse needs, such as the elderly who require accessibility features (e.g., screen readers, voice commands, and color blindness support).
- **Doctor:** Healthcare providers who use the system to conduct video consultations, issue prescriptions, update patient medical records, and manage patient care remotely. They are responsible for delivering treatment and advice, accessing patient data securely, and communicating with patients through the system.

Supporting Actors

- **System Administrator:** Responsible for managing the technical infrastructure of the system, ensuring it is available 24/7. They oversee security protocols, manage system performance (including load balancing and scalability), and conduct regular security audits and penetration testing. They ensure compliance with data protection regulations and standards such as the PDPA (Personal Data Protection Act) [5] and ISO 27001.
- **Hospital/Clinic Staff:** Support personnel who assist patients with inquiries, manage appointment bookings, and facilitate communication between patients and doctors. They help ensure a seamless user experience and handle issues related to patient data or system functionality, ensuring smooth system operations.
- **Security Manager:** A specialized role responsible for ensuring the system's security measures are properly implemented. They manage security tools such as Endpoint Detection and Response (EDR) [2], Security Information and Event Management (SIEM) [3], and Security Orchestration, Automation, and Response (SOAR) [4]. They are responsible for encryption standards, multi-factor authentication (MFA), and compliance with regulations such as HIPAA, the PDPA [5], and the Thai Medical Act [6].

- **Payment Gateway:** Facilitates secure financial transactions for medical services and medication purchases. It ensures that payments are processed securely using encryption and compliance with legal and financial standards, such as PCI-DSS, to protect patient financial data.
- **Credit Card Processors:** These systems handle the secure processing of credit card payments for medical services and medications. They work alongside payment gateways to ensure that credit card transactions are completed securely and comply with financial standards, including encryption and fraud prevention protocols.
- **Mobile Banking Systems:** These systems integrate with mobile banking applications to allow patients to pay for services and medications securely through their mobile devices. They ensure compliance with security measures, including encryption and multi-factor authentication (MFA), to protect patient financial information during transactions.
- **Regulatory Authority:** External entities ensuring that the system complies with local and international regulations, such as the Personal Data Protection Act (PDPA), the Thai Medical Act [6], and HIPAA (Health Insurance Portability and Accountability Act). They may require periodic audits, certifications (e.g., ISO 27001), and compliance reporting.
- **Cloud Service Provider:** Provides the cloud-based infrastructure that supports the system's high availability, scalability, and performance. They manage concurrent transactions, ensuring the system remains operational 24/7 with minimal downtime. Cloud service providers are critical for maintaining system uptime and reliability.

Use Case Diagram



Link: [Use Case Diagram](#)

Use Case Narrative

Use Case Name	Make Remote Appointments
Goal in Context	Enable patients to schedule, reschedule, or cancel medical appointments remotely through the Hospital CRM system.
Primary Actor	Patient
Secondary Actor	• Doctor • Hospital Staff
Precondition	• The patient must be registered and logged into the system. • Available appointment slots must exist for the selected doctor or clinic.
Trigger	The patient initiates the appointment booking process by selecting the "Make Appointment" option on the CRM system.
Scenario	1. The patient logs into the Hospital CRM system. 2. The patient navigates to the “Make an Appointment” section. 3. The system displays a list of available doctors and appointment slots. 4. The patient selects a doctor based on specialization and availability. 5. The system prompts the patient to choose a date and time for the appointment. 6. The patient confirms the chosen date and time. 7. The system verifies the availability of the selected slot. 8. The system sends an appointment confirmation notification to the patient via email/SMS. 9. The doctor and relevant hospital staff are notified of the new appointment. 10. The system sets reminders for the appointment (e.g., 24 hours and 1 hour prior).
Exceptions	1. No available time slots – The system prompts the patient to select a different date or doctor. 2. Incomplete patient information – The system requires the patient to update their profile or medical history before confirming the appointment. 3. System error during appointment booking – The system

	displays an error message and prompts the user to retry.
Post-condition	• The appointment is successfully booked and confirmed. • Notifications and reminders are sent to both the patient and the doctor. • The appointment is added to the patient's and doctor's schedules in the system.

Use Case Name	Receive Appointment Reminders
Goal in Context	Ensure patients receive timely reminders for their scheduled appointments to reduce missed consultations.
Primary Actor	Patient
Secondary Actor	• Doctor • Hospital Staff
Precondition	• The patient has a confirmed appointment (remote or in-person) scheduled in the system. • The patient has provided contact information (email, phone number, or preferred communication method).
Trigger	The scheduled appointment time is approaching (e.g., 24 hours before the appointment).
Scenario	1. The system checks for upcoming appointments within the predefined reminder period (e.g., 24 hours and 1 hour before). 2. The system sends an automated reminder to the patient via the patient's preferred communication channel (email, SMS, or in-app notification). 3. The system provides details such as the appointment date, time, doctor's name, and any instructions (e.g., preparing for a remote consultation). 4. The system also sends a reminder to the doctor about the scheduled appointment. 5. If the patient confirms or cancels via the reminder message, the system updates the appointment status accordingly. 6. The system logs the reminder notification in the patient's record.
Exceptions	1. Invalid contact details – If the patient's contact information is outdated or incorrect, the system flags it and prompts the patient to update their contact information.

	2. Patient does not receive the reminder (e.g., due to email or SMS server issues) – The system attempts to resend the reminder or notify hospital staff if delivery repeatedly fails.
Post-condition	• The patient is reminded of their upcoming appointment, reducing the likelihood of missed consultations. • The system logs the reminder event, and any confirmations or changes are updated in the patient's record.

Use Case Name	Manage Appointment Schedules
Goal in Context	Allow patients to manage (view, reschedule, or cancel) their appointments efficiently through the hospital CRM system.
Primary Actor	Patient
Secondary Actor	• Doctor • Hospital Staff
Precondition	• The patient must have an active account in the system. • The patient has scheduled at least one upcoming appointment.
Trigger	The patient logs into the system and accesses their appointment schedule.
Scenario	1. The patient logs into the hospital CRM system. 2. The patient navigates to the "Appointments" section, where they can view a list of scheduled appointments. 3. The patient selects an appointment they wish to manage. 4. The system provides options to: ○ Reschedule: The patient selects a new date and time from the available options, and the system checks for the doctor's availability. ○ Cancel: The patient confirms the cancellation, and the system updates the doctor's schedule. 5. The system updates the appointment status (rescheduled or canceled) and notifies the doctor of the change. 6. If the appointment is rescheduled, the system sends a confirmation message with the new details to the patient. 7. The system logs the action in the patient's record.
Exceptions	1. No available slots – If the patient attempts to reschedule but no slots are available, the system provides alternative options or

	prompts the patient to contact hospital staff. 2. System outage – If the system is unavailable, the patient is advised to contact hospital staff for manual appointment management. 3. Deadline exceeded – If the patient tries to cancel an appointment after the allowed time window (e.g., within 24 hours of the appointment), the system blocks the action and prompts the patient to contact hospital staff.
Post-condition	• The patient’s appointment schedule is updated according to their changes (rescheduled or canceled). • The system notifies the doctor of any changes to their schedule.

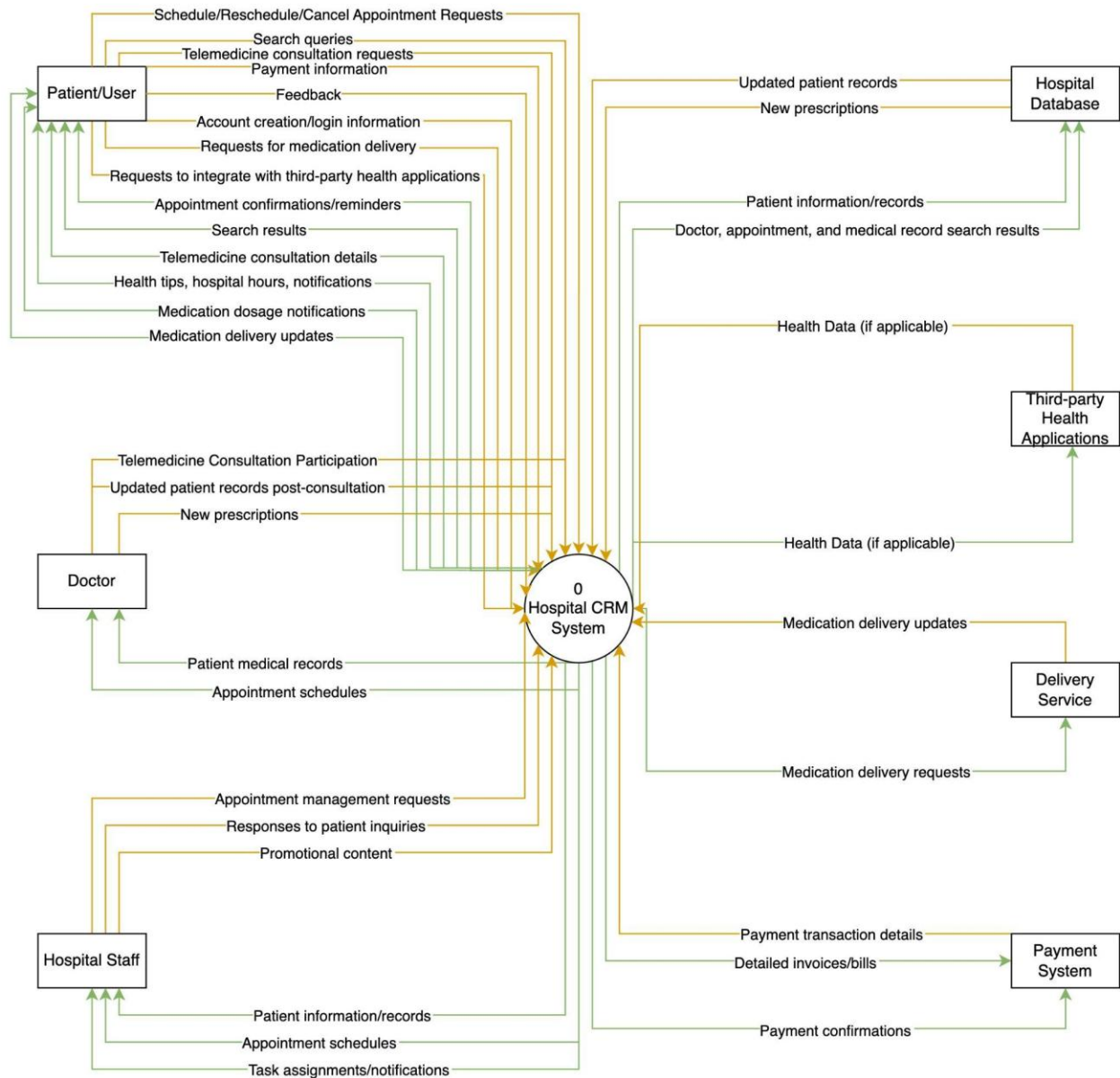
Use Case Name	Process Secure Payments
Goal in Context	Allow patients to securely pay for medical services and medications via the CRM system, supporting both credit card and mobile banking payment options.
Primary Actor	Patient
Secondary Actor	• Payment Gateway • Hospital Staff • System Administrator
Precondition	• The patient has received medical services or purchased medication, and an invoice has been generated. • The patient has an active account in the CRM system.
Trigger	The patient receives an invoice and initiates the payment process.
Scenario	1. The patient logs into the CRM system. 2. The system displays the patient’s outstanding invoice(s). 3. The patient selects “Make Payment” and chooses a payment method (credit card or mobile banking). 4. The system connects to the secure payment gateway. 5. The patient enters payment details (credit card or mobile banking credentials). 6. The payment gateway processes the transaction. 7. The system verifies the payment and updates the hospital’s billing records. 8. The system sends a payment confirmation to the patient via email or notification.

	9. The hospital staff is notified of the successful payment.
Exceptions	1. If the payment is declined or fails, the system notifies the patient and prompts them to try another payment method. 2. If the payment gateway is down, the system notifies the patient of a delay and attempts to retry the transaction later. 3. If there is insufficient balance in the patient's account, the system alerts the patient to retry when funds are available.
Post-condition	• The payment is processed securely, and the patient receives a receipt • The hospital's billing system is updated with the completed transaction.

Use Case Name	Update Patient Records Post-Consultation
Goal in Context	Allow doctors to update and manage patient records after consultations, ensuring accurate and up-to-date medical information.
Primary Actor	Doctor
Secondary Actor	• Patient • System Administrator • Hospital Staff
Precondition	• The doctor has completed a consultation with the patient. • The patient's medical records exist in the system.
Trigger	The doctor needs to update the patient's medical records after a consultation.
Scenario	1. The doctor logs into the CRM system. 2. The doctor selects the patient's medical record. 3. The doctor updates the medical notes, diagnosis, or treatment plan. 4. The doctor saves the updated information, and the system securely stores the new data. 5. The system notifies the patient that their medical records have been updated. 6. The patient can access their updated records through the CRM system.

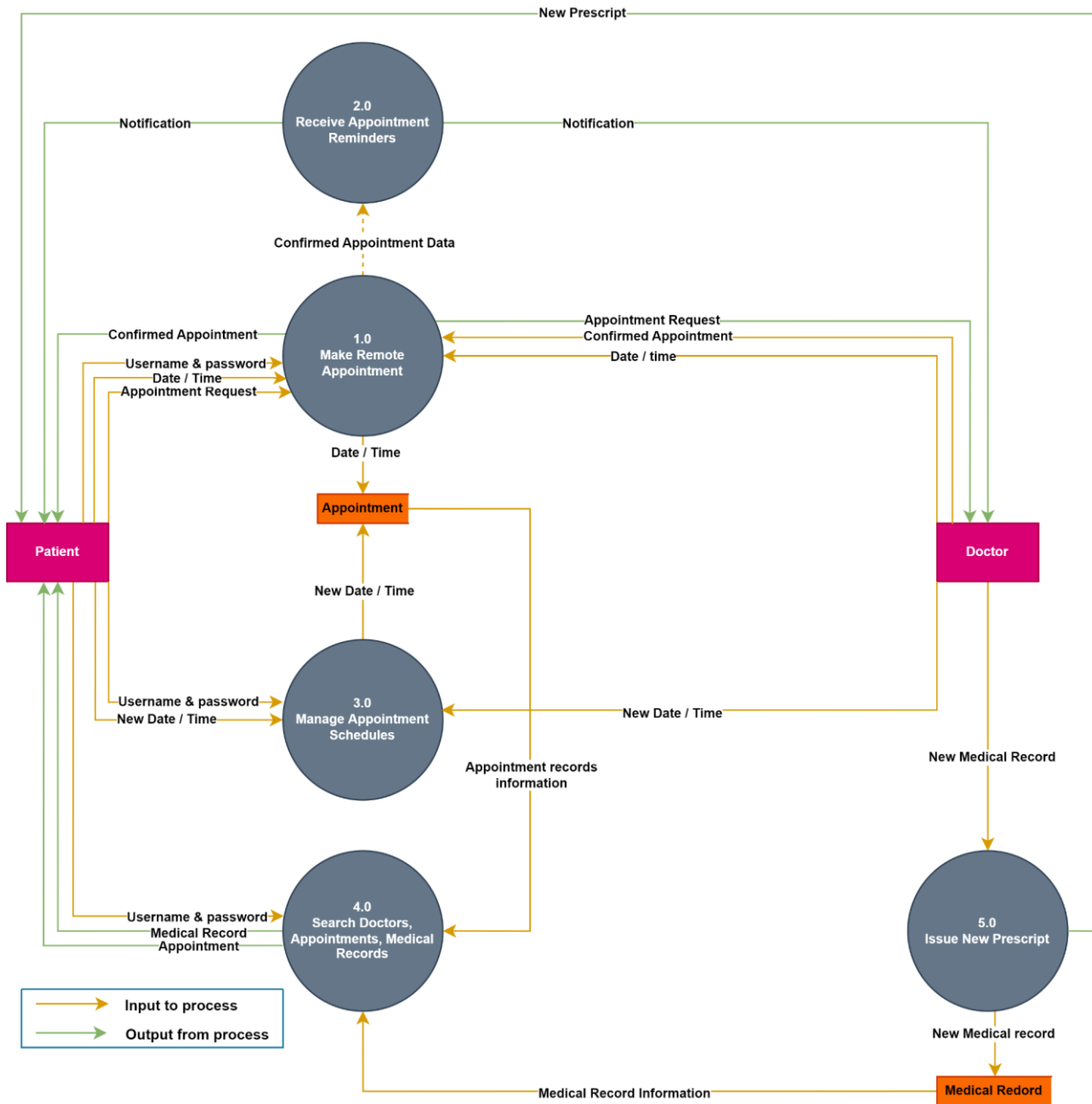
Exceptions	1. If the doctor's session times out due to inactivity, the system prompts the doctor to log in again and resume the update. 2. If the system detects a data conflict (e.g., simultaneous updates by hospital staff), it prompts the doctor to review the changes before saving. 3. If the system encounters a failure in saving the updates, it retries automatically and logs an alert for the system administrator.
Post-condition	• The patient's medical records are updated and securely stored in the system. • The patient is notified of the updates and can access the new information.

Context Diagram (DFD Level 0)



Link: [DFD Level 0](#)

DFD Level 1



Link: [DFD level 1](#)

References

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